Prostart: Democratizing Sprint Training with a Low-Cost Reaction-Time and Photo-Finish System for Track & Field

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Abstract

Sprint races are decided by milliseconds, yet the systems that time reaction and finish are costly [1] and stay with federations and large clubs, while everyone else times by hand. We present Prostart: a start unit that mounts on any starting block, and a phone app that turns a smartphone into a photo finish. The unit, an ESP32-C3 with a BMI270 accelerometer sampled at 804 Hz on a custom PCB in a 3D-printed box, gives the start commands, finds the push-off with a threshold-independent two-stage onset picker, and sends the "go" instant over Bluetooth Low Energy to the phone, which times the chest crossing the line with an on-device pose model, a synchronized clock and sub-frame interpolation. Against 240 fps video with an LED marking "go", reaction times agreed within about 8 ms, the uncertainty of the video reference itself; finish times agreed with a 60 fps reference within 10 ms, a third of the one-frame (33 ms at 30 fps) target we set, whereas hand timing read about 0.24 s fast.

1 Introduction

In competitive sprinting, athletes train for years to gain a few hundredths of a second: Noah Lyles won the Paris 2024 Olympic 100 m final by five thousandths [2]. Reaction times differ by several hundredths between elite sprinters [3], so part of the winning margin is decided before the athlete has even moved. Reaction time runs from the starting signal to the first force on the block; under 0.100 s is a false start, a limit based on the minimum auditory reaction time [4], and the fastest valid Olympic reactions lie just above it [3, 5]. The systems that measure it are installed at certified meets [6], and even a single-block training version lists at about US$1,600 [1]. Coaches therefore time by hand, which reads about 0.24 s faster than electronic timing [7]: an error larger than the gains an athlete trains for.

Photocells start when the torso breaks a beam, so their times exclude the reaction [8]. FinishLynx ReacTime measures reaction to 1/1000 s from a sensor on every block [6]; its patent detects the onset when acceleration crosses a threshold [9], which by construction reports it late. Omega samples the force on the footrest at 4 kHz and pairs it with a finish camera of up to 40,000 images per second [10], and earlier student work used load cells, robust in 18 of 20 starts but without a timing-accuracy characterization [11]. All of these are either costly or built into the block, leaving a gap for a low-cost, portable system with a documented error budget.

This paper contributes (i) a start unit on one microcontroller, from schematic to PCB to printed enclosure, whose onset picker is independent of its threshold; (ii) a Flutter app that mirrors the start and turns the phone camera into a photo finish, synchronized to the unit over BLE; and (iii) a validation of both against high-speed video, with every error term named. The unit started as a breadboard prototype (Appendix B) and is now a complete system: the unit in its printed box and the app (Figure 1).

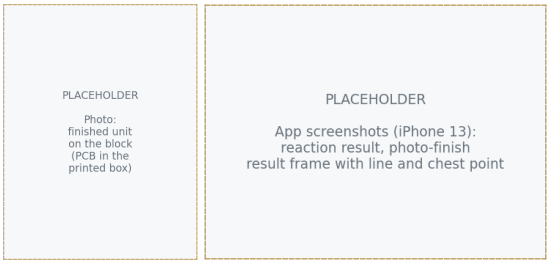


Figure 1. The complete system: the start unit in its printed box on the block (left) and the app (right).

2 System design

Architecture. Two devices and one radio link (Figure 2). The start unit is the only judge: it sounds the commands, stamps them and the push-off on its own clock, and decides the verdict. The phone, standing at the finish, mirrors the start and times the finish; nothing it sends can change a start. A single ESP32-C3 reads the accelerometer and runs the radio. Each reading is timestamped in an interrupt and buffered in the sensor's FIFO, so Bluetooth traffic can delay it but never shift or lose it. A direct BLE link covers about 30 m, enough for start and acceleration work; for a full 100 m we plan a second ESP32-C3 at the finish, linked to the start unit over long-range ESP-NOW, which we field-tested beyond 100 m, and to the phone over BLE.

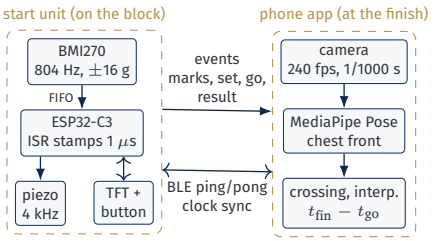


Figure 2. System overview. The start unit judges the start; the phone receives the "go" instant on its own clock and times the finish.

Hardware. A two-layer PCB (65.5×34 mm, Appendix B) carries an ESP32-C3 SuperMini, a BMI270 accelerometer breakout on I^2C at 400 kHz with its data-ready line on an interrupt pin, a 1.77" TFT on SPI, a push button, and a piezo driven in antiphase from two pins, which doubles its swing without an amplifier; 10 μF and 100 nF decouple the 3.3 V rail. The enclosure, 75×99.5×25 mm, was designed around the board and printed in three parts (Figure 3): a box with standoffs for the PCB, a lid with a window for the display and openings for the buzzer, and a button cap.

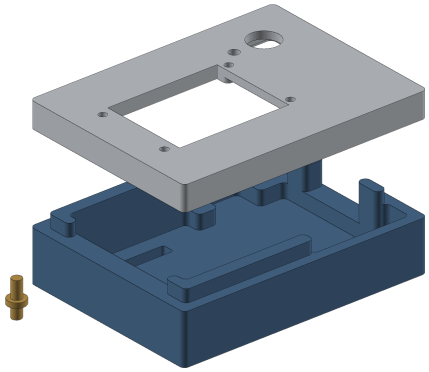


Figure 3. The printed enclosure, exploded, rendered from its STL files.

Start sequence. A press starts a 3 s countdown, then "on your marks"; "set" follows 20–25 s later. The unit then waits until the sprinter is measured still (horizontal acceleration under 15 mg for 200 ms) and fires "go" a random 0.7–1.5 s later, so the rhythm cannot be learned. Each beep is a 4 kHz, 100 ms tone started by two register writes and stamped at that instant, so no software latency enters the reaction time.

Onset detection. The BMI270 runs at 800 Hz nominal (803.6 Hz measured), ±16 g. Gravity is estimated at rest as a vector, so every sample splits into vertical and horizontal parts whatever the mounting angle. An STA/LTA trigger [12] (15/800 ms windows, ratio 6, 20 mg floor) decides that the athlete moved; Maeda's AIC picker [13] then finds, from 150 ms before to 50 ms after the trigger, the sample that best splits a quiet segment from an active one, and reports it as the onset. Because this second step has no threshold, the onset does not move with the trigger level: across 79 starts and thresholds of 10–50 mg it stayed on the same sample in 70 starts and within one in 74, while a threshold-only detector drifted by a median 9.3 ms (Appendix D). Runs are stored on flash and can be exported as CSV.

App and clock synchronization. The Flutter app (iOS and Android) connects to the unit's GATT service. In *Reaction* mode it mirrors the unit's phases and shows its result. In *Photofinish* mode the board's "go" instant has to be expressed on the phone's clock. Between "on your marks" and "set" the phone pings the board; the board answers with its receive and send times. Over a 10 s window the app keeps the minimum of each one-way difference separately,

$$a = \min(t_{\mathrm{rx}} - t_1), \quad b = \min(t_4 - t_{\mathrm{tx}}), \quad \theta = \tfrac{b-a}{2},$$

where t_1, t_4 are the phone's send and receive stamps and θ is the phone–board offset. Minimizing each direction on its own matters on BLE: a reply waits for the next connection event, so a single round trip is asymmetric by up to one interval (15–30 ms on iOS), while each direction's minimum is a message that just made its event [14]. The residual error is bounded by half the minimum round trip, and drift (20–50 ppm) adds under 0.5 ms over the window.

AI photo finish. The phone stands in line with the finish, and the user drags a line onto the painted one. The camera runs at its highest frame rate (240 fps on an iPhone 13), with stabilization off and a short exposure (1/1000 s) against motion blur; every frame keeps the time it was captured, on the same clock the synchronization uses. From "go" the frames are kept in memory, and when a body enters a narrow band around the line, MediaPipe Pose [15] analyses the pass. The rules time the torso [16], but pose landmarks are joints inside the body, so the torso centre crosses about 20 ms after the chest. We therefore take the chest front from the body silhouette: between shoulders and hips, with the arm removed, the frontmost edge of each row, and the 80th percentile of these edges, so a stray forearm cannot move it. Between the last frame before the line and the first after it, interpolation gives the crossing instant, and a line fit over ±160 ms smooths per-frame noise. The time is shown to the millisecond and, as the rules prescribe, rounded up to the hundredth [16]. The detector was first written in Python and then ported to the app; on the same clips the two agree within a few ms (1.993 vs. 1.994 s at 30 fps; 0.864 vs. 0.862 s at 240 fps).

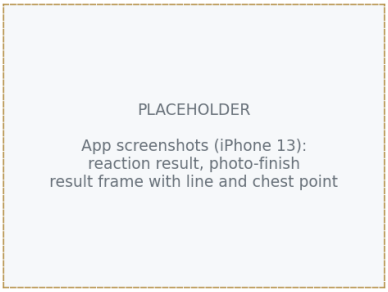


Figure 4. The app: reaction result mirrored from the unit (left) and photo-finish result with the line and the timed chest point (right).

3 Evaluation

Timing path. On the PCB the unit samples at 803.6 Hz with no gaps and no dropped samples,

a maximum interval of 1.25 ms, and 0.55 μs timestamp jitter. Re-running the detector offline on every capture gives the board's onset on the same sample (e.g. 163.8 ms). The unit's own resolution is therefore one sample, 1.2 ms. Figure 5 shows what the detector sees in a block start.

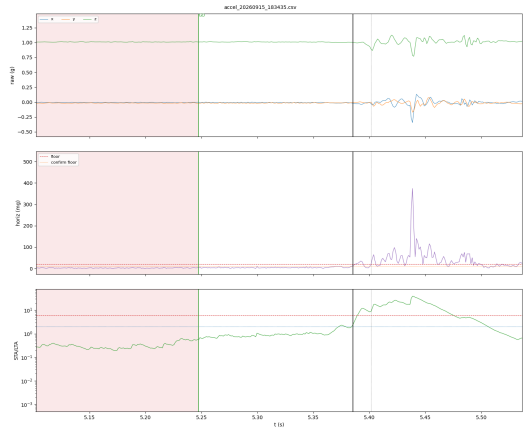


Figure 5. A block start recorded with the first prototype: raw acceleration, horizontal magnitude and STA/LTA ratio after "go" (green), with the AIC onset (black) at 138 ms, 16.2 ms before the threshold-only onset (dotted).

Reaction time vs. 240 fps video. For ground truth we added an LED that the firmware lights at the "go" instant, and filmed the block at 240 fps (4.2 ms per frame) in **N** starts. The first lit frame gives "go" exactly to the frame; the push-off is the first frame in which the athlete visibly moves, chosen by eye. Board and video agreed within about 8 ms (Figure 6). This is the resolution of the reference, not of the unit: one frame alone is 4.2 ms, and the eye picks the first visible motion of the body, whereas the accelerometer feels the first force on the block. Two constant terms remain unmeasured, the BMI270 filter delay and the sound's travel to the athlete (2.9 ms per metre). We are confident the unit stays within its 1.2 ms sample; showing it directly needs a 1000 fps camera, our next step.



Figure 6. Reaction time, board vs. 240 fps video, one point per start: difference against mean.

Photo finish. The finish time combines the sync, the frame timing and the detector, so we validated it end to end. Hand timing on the same runs read about 0.24 s fast, matching the literature [7]. Against a 60 fps reference video read frame by frame, the app's time differed by 10 ms, less than one 60 fps frame (16.7 ms) and well within the 33 ms (one frame at 30 fps) we set as the target at half-time: interpolating between frames gives a sub-frame time (Figure 7). Figure 8 puts every error in this paper on one scale.

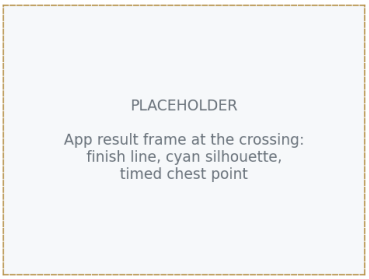


Figure 7. The app's result frame at the crossing: finish line, body silhouette (cyan) and the timed chest point.

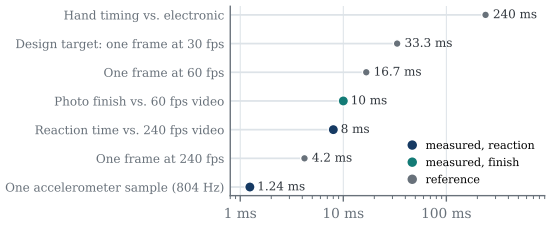


Figure 8. Every timing error in this paper on one log axis: measured disagreements (filled) against frame and sample intervals and hand timing.

Limitations. Besides the two constant delays above, the direct BLE link reaches about 30 m, so a full 100 m needs the relay planned in Section 2. In dim light the camera cannot shorten the exposure below about 1/250 s, and motion blur grows. An arm swung across the chest can hide the chest edge from the silhouette, and with two athletes in the frame the detector times the larger torso, so the app is meant for one athlete per run. Finally, the phone must stand in line with the finish: off the line, parallax shifts the crossing.

4 Conclusions

We built a start unit on one microcontroller, from PCB to printed box, and a phone app that together time a sprinter's reaction and finish, for parts costing **~X SEK**. Against high-speed video both agree within the reference's own uncertainty, 8 ms for reaction and 10 ms for the finish, against 240 ms for a stopwatch. Next, a 1000 fps camera will check the unit's 1.2 ms resolution directly and measure the constant delays of the sensor and the sound.

References

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A Source code

Firmware, app, photo-finish reference, analysis tools, PCB project and enclosure files: https://github.com/lorenzogalli-dev/Reaction-Time-System. The firmware is in `Arduino/AlgorithmRealTimeESP32/`, the app in `prostart/`, the KiCad project in `Hardware/PCB_ProStart_POC/`, the STL files in `Hardware/Enclosure/`.

B Hardware

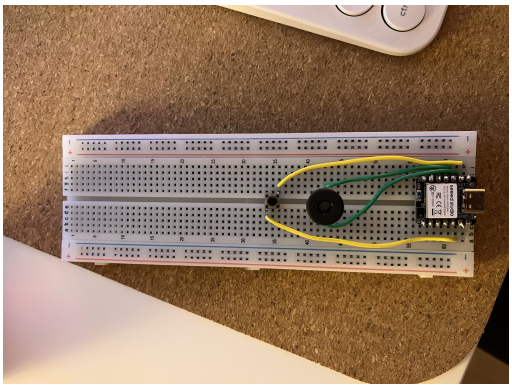


Figure 9. The first prototype on a breadboard: microcontroller, button and buzzer.

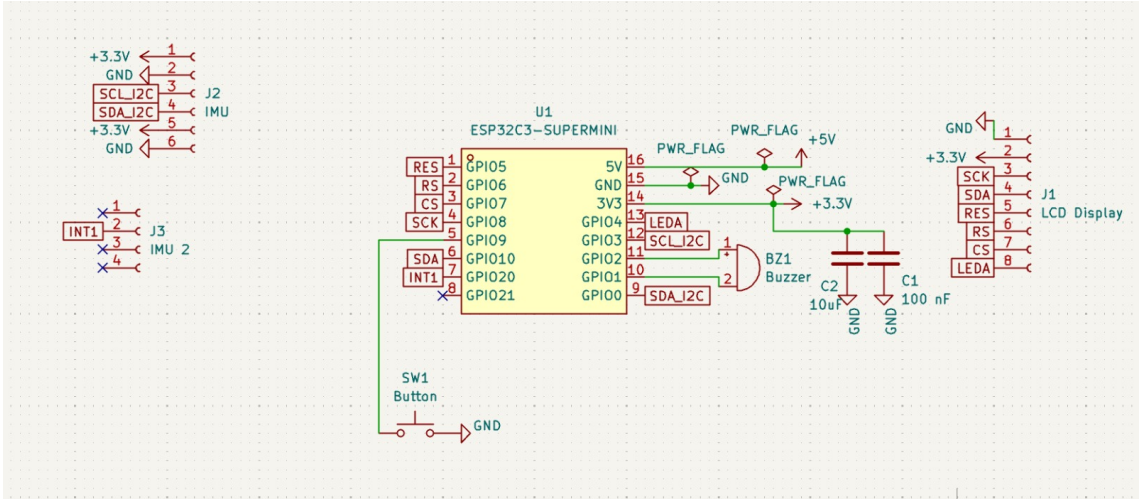


Figure 10. Schematic of the start unit. J2/J3: BMI270 breakout (I^2C, INT1); J1: TFT display; BZ1: piezo between GPIO1 and GPIO2.

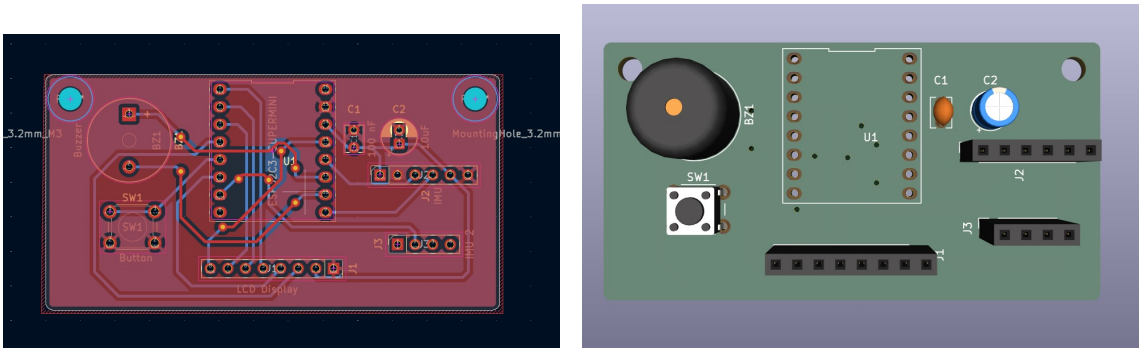


Figure 11. Two-layer PCB, 65.5×34 mm: routed layout (left) and 3D view (right).

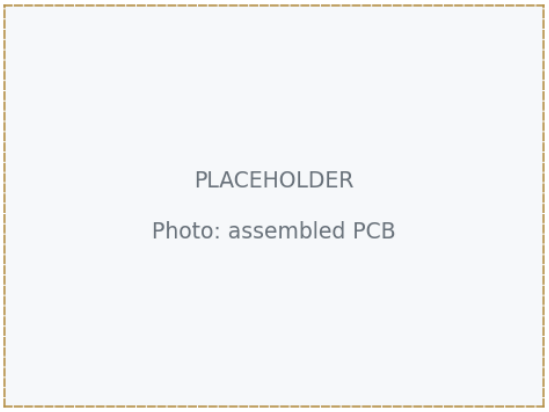


Figure 12. The assembled PCB.

Ref.	Part	Connection
U1	ESP32-C3 SuperMini	–
J2, J3	BMI270 breakout [17]	SDA GPIO0, SCL GPIO3, INT1 GPIO20
J1	1.77” ST7735 TFT, 128×160	SCK 8, MOSI 10, CS 7, DC 6, RES 5, BL 4
SW1	6 mm push button	GPIO9 to GND
BZ1	12 mm piezo buzzer	GPIO1 / GPIO2, antiphase
C1, C2	100 nF, 10 μF	3.3 V rail

Table 1. Bill of materials and pin map.

C Firmware parameters and BLE protocol

Parameter	Value
Sampling	BMI270, 800 Hz nominal (803.6 Hz measured), ±16 g, 0.488 mg/LSB, FIFO 1.28 s, data-ready stamped in the ISR (1 μs clock)
Sequence	countdown 3 s; marks → set 20–25 s; arm when <15 mg for 200 ms (at least 800 ms after set, at most 4 s); go 0.7–1.5 s after arming
Beep	4 kHz, 100 ms, two register writes, stamped at the write
STA/LTA	15 ms / 800 ms, on at ratio 6, off at 2, 20 mg floor, 15 ms confirmation
AIC window	150 ms before, 50 ms after the trigger
False start	onset < 100 ms after go
Storage	25 runs on internal flash, CSV export over USB

Table 2. Start-unit parameters.

Characteristic	Content
`event` (notify)	type, counter (a gap reveals a lost event); countdown digit; marks/set/go board time (μs); result: verdict, reaction time, onset and go times
`ping` (write)	16-bit sequence number from the phone
`pong` (notify)	sequence, board receive time, board send time; answered only before "set"

Table 3. BLE GATT service of the start unit. Board times are 32-bit microseconds, unwrapped by the app.

D Threshold independence of the onset

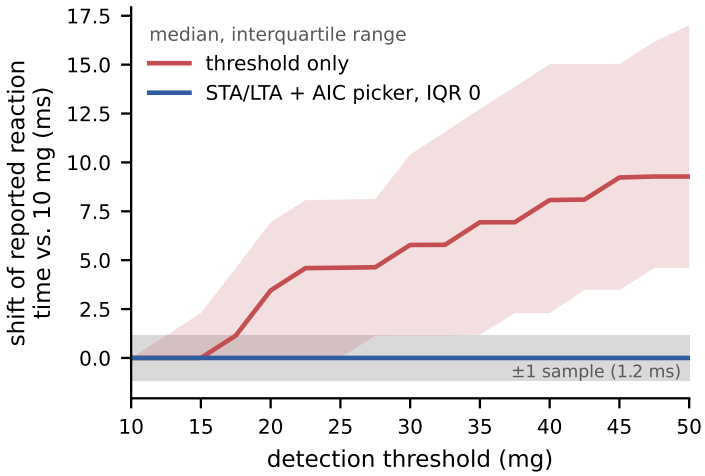


Figure 13. Threshold sweep over 79 block starts recorded with the half-time unit (10–50 mg): the AIC onset stays fixed while threshold-only detection drifts (median 9.3 ms).